

Introduction to MDX Queries

What Is MDX?

- MDX is Multi Dimensional EXpressions
- MDX is the syntax for querying an Analysis Services database
- MDX is part of the OLE DB for OLAP spec
- MDX is the key for all advanced analytical capabilities of Analysis Services

Comparison To SQL

<i>SQL Construct</i>	<i>OLAP construct</i>
SELECT...	SELECT... (MDX)
CREATE...	DSO object model
DROP...	
INSERT...	
DELETE...	

MDX Basics

- MDX allows easy navigation in the multi dimensional space
- It “understands” the MD concepts of cube, dimension, level, member and cell
- It is used for
 - Queries – full statements (SELECT... FROM)
 - Business modeling – defining calculated members using MDX Expressions – not a full statement

MDX Queries vs. MDX Expressions

- MDX Queries
 - Full statements (SELECT...FROM)
 - Usually generated by a query tools and applications such as Excel
 - MDX Sample App deals in queries
- MDX Expressions
 - Partial MDX statements
 - Define a calculated member, or a set, or member properties, etc.
 - Returns a single value (which may be a set)

MDX Myth

“Only developers need to know MDX”

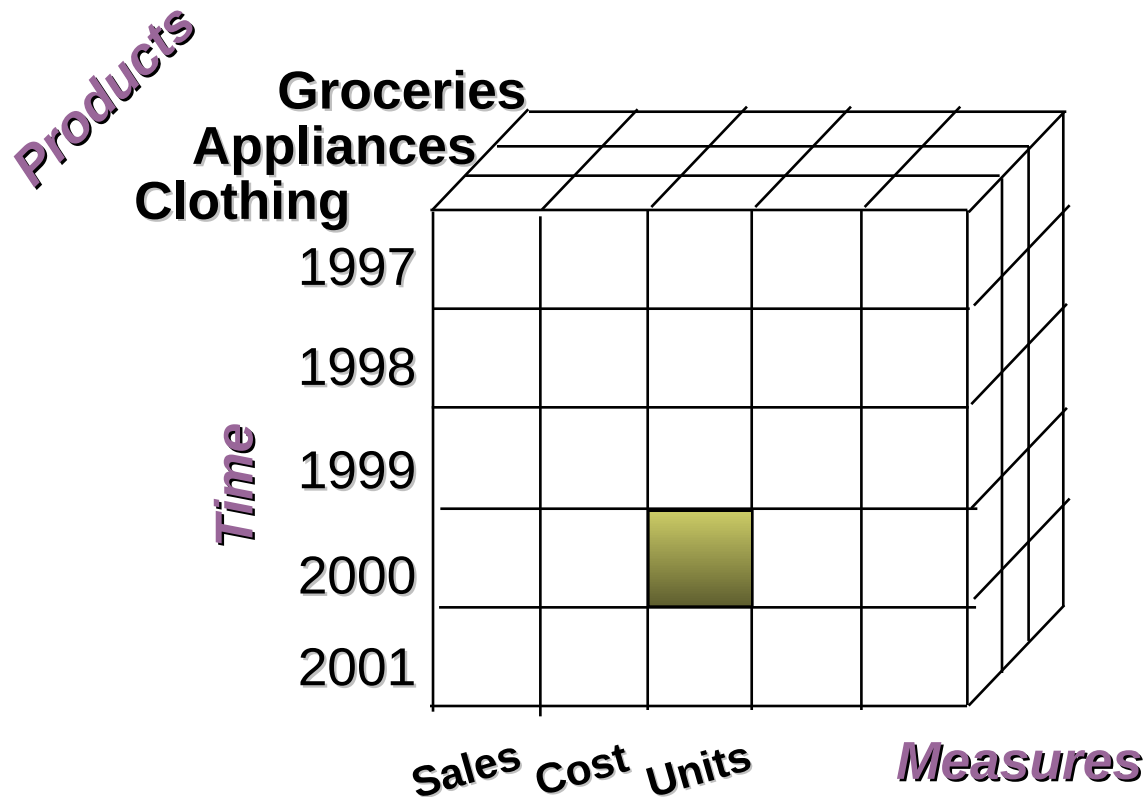
Fact: MDX is used everywhere:

- **Calculated members**
- **Security settings**
- **Custom member rollup**
- **Custom level rollup**
- **Actions**
- **Define named “Sets”**
- **Calculated cells**

MDX Constructs

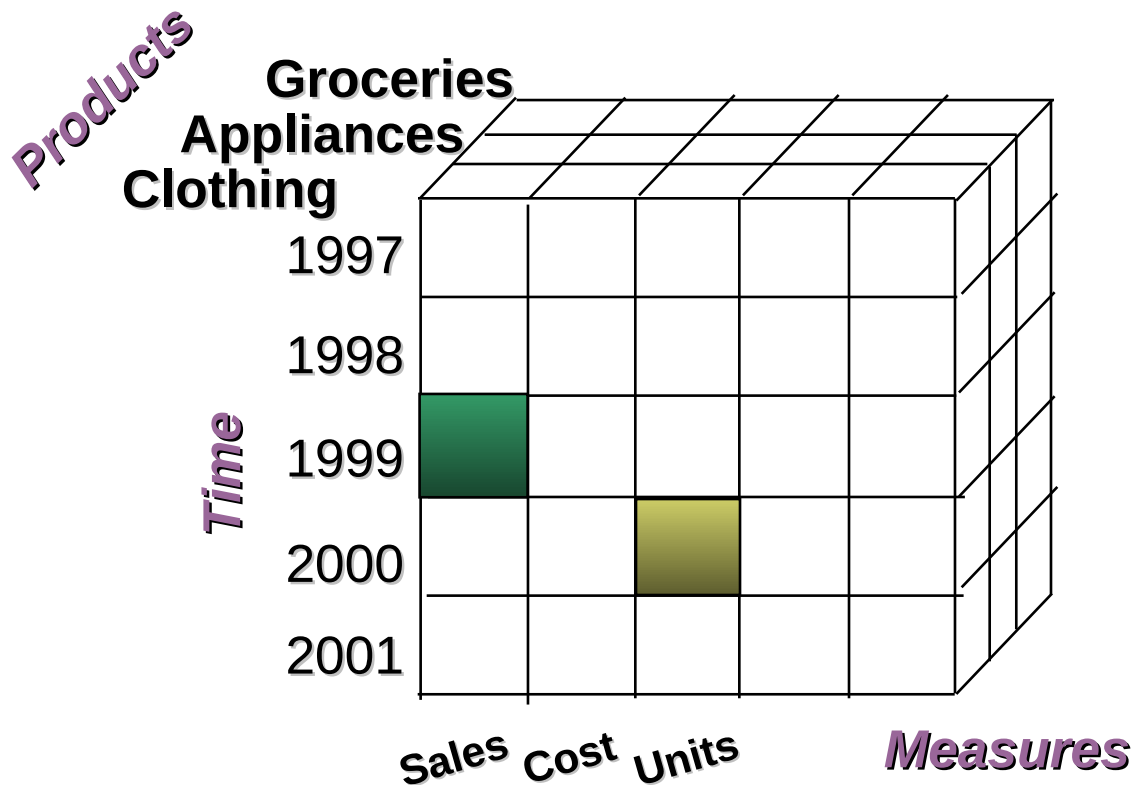
- Members: an item in a hierarchy
 - [John Doe]
 - [2001]
 - [2001].[Q1].[Jan]
- Tuple: an intersection of 2 or more members
 - ([Product].[Drink].[Beverages], [Customers].[USA])
 - ([Product].[Non-Consumable], [2001])
- Sets: a group of tuples or members
 - {[John Doe], [Jane Doe]}
 - { ([Non-Consumable], USA), (Beverages, Mexico) }
 - [2001].Children
 - TopCount(Store.[Store Name].
Members, 10, Sales)

Every Cell Has A Name...



Every Cell Has A Name...

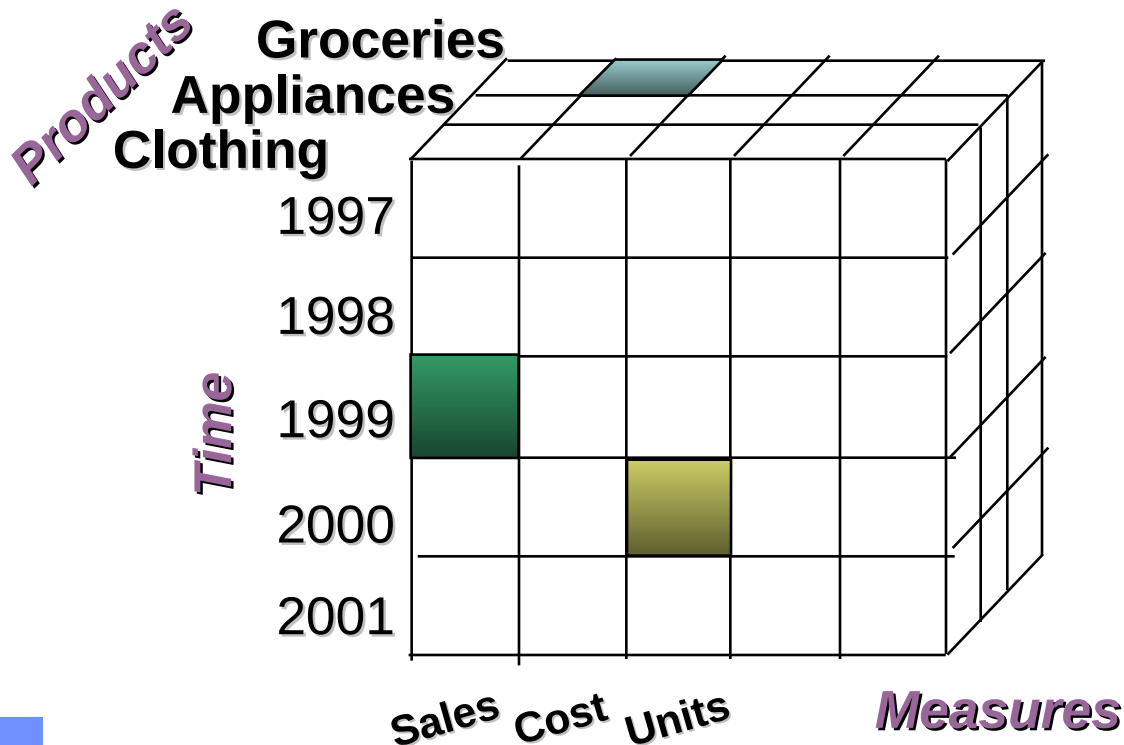
(Products.Clothing, Measures.Units, Time.[2000])



Every Cell Has A Name...

(Products.Clothing, Measures.Units, Time.[2000])

(Products.Clothing, Measures.Sales, Time.[1999])

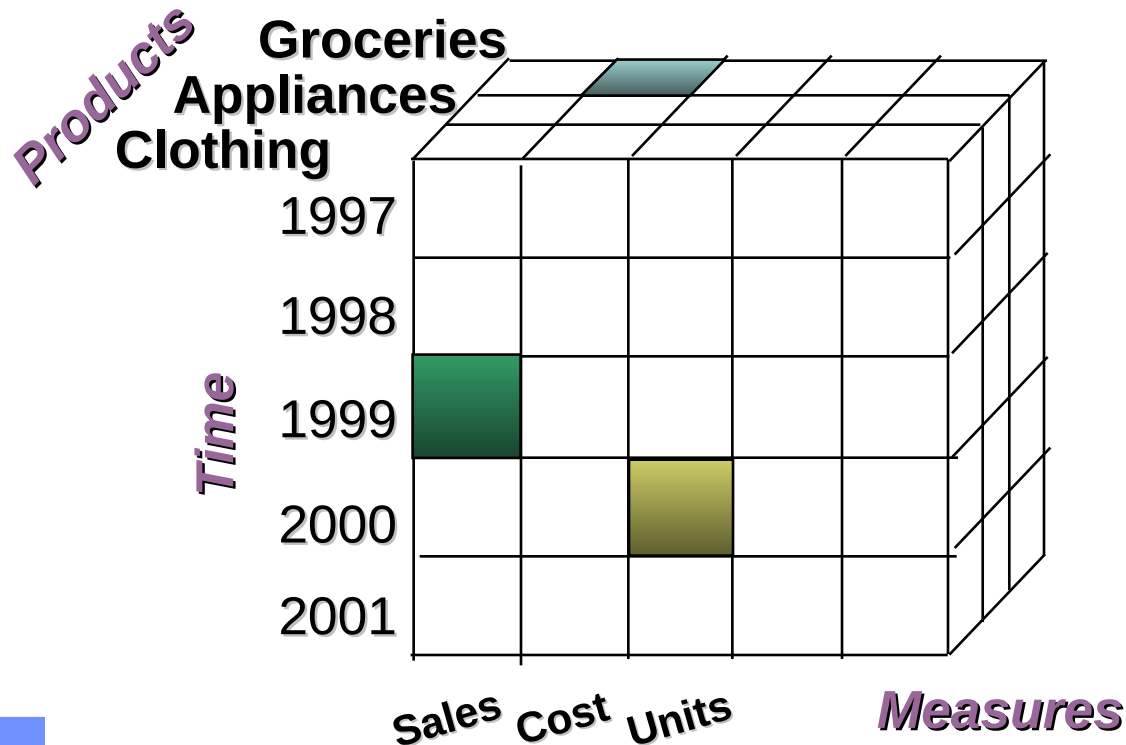


Every Cell Has A Name...

(Products.Clothing, Measures.Units, Time.[2000])

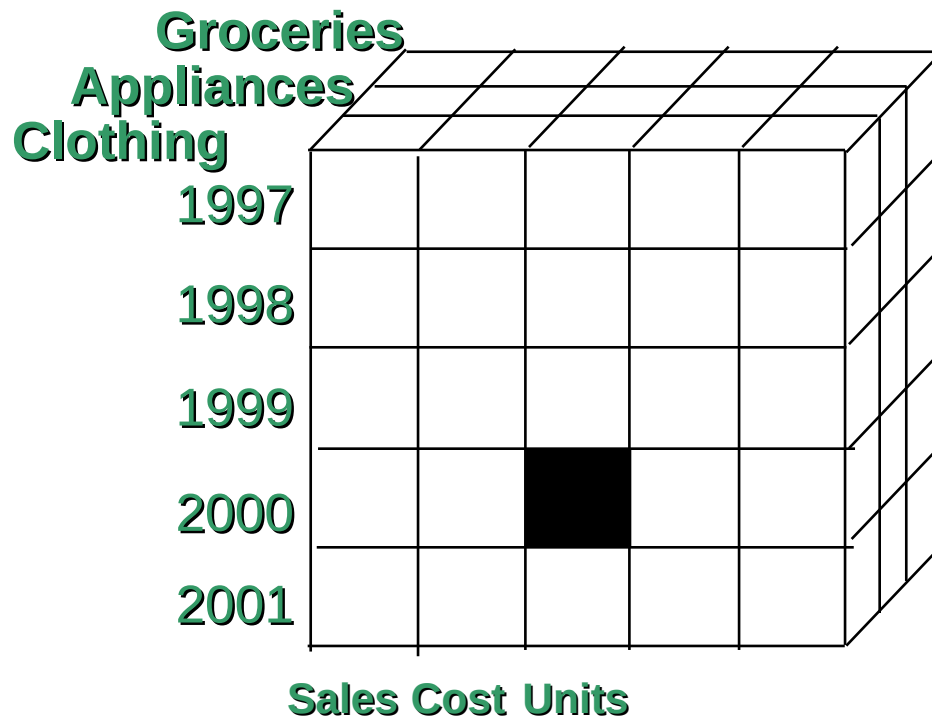
(Products.Clothing, Measures.Sales, Time.[1999])

(Products.Groceries, Measures.Cost, Time.Year.[1997])

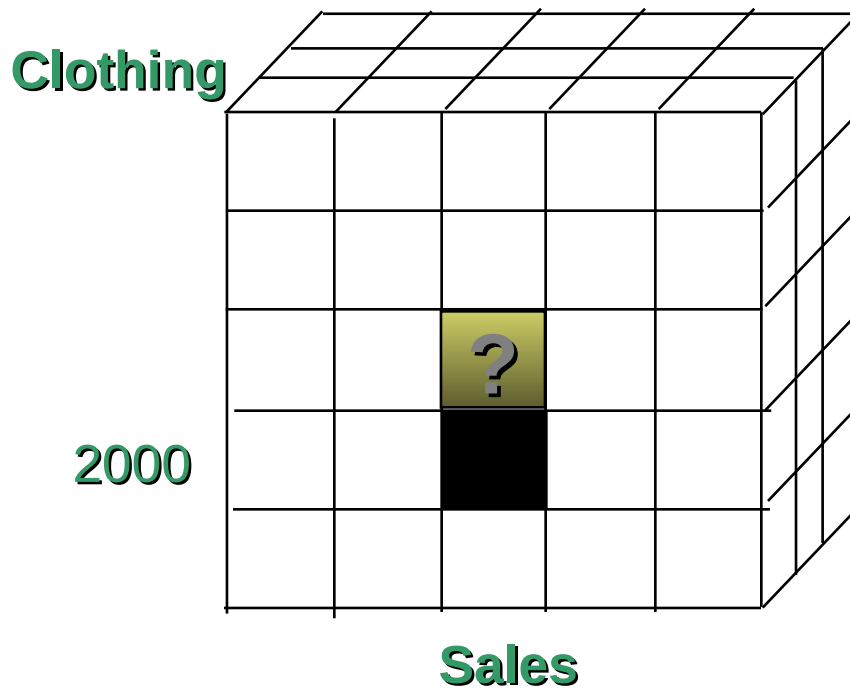


The Current Cell: Currentmember

**(Products.Clothing, Measures.Units, Time.[2000]) =
(Products.CurrentMember,
Measures.CurrentMember,
Time.CurrentMember)**

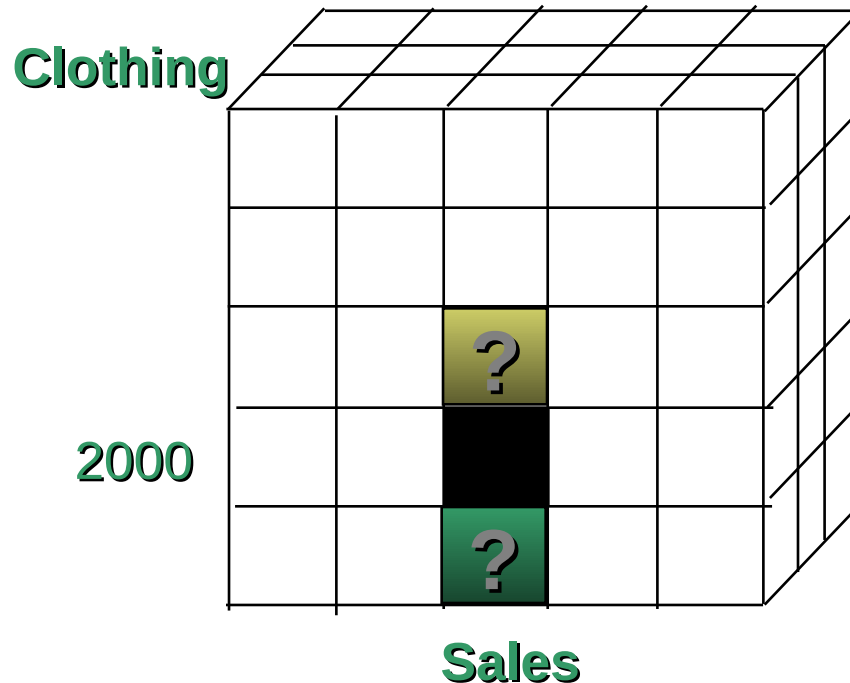


Naming Cells With Relative References...



Naming Cells With Relative References...

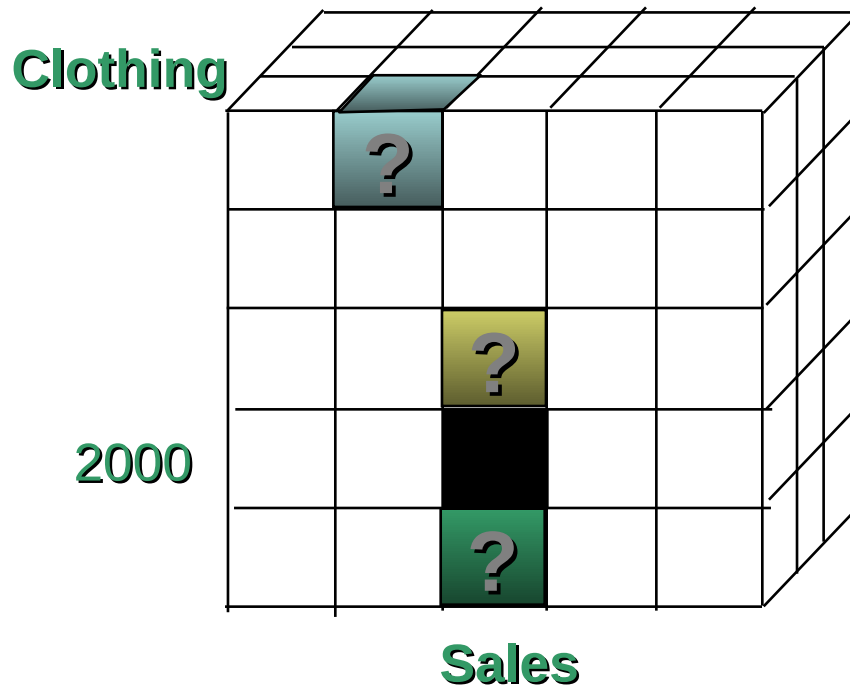
(Products.Clothing, Measures.Sales, Time.[2000].PrevMember)



Naming Cells With Relative References...

(Products.Clothing, Measures.Sales, Time.[2000].PrevMember)

(Products.Clothing, Measures.Sales, Time.[2000].NextMember)

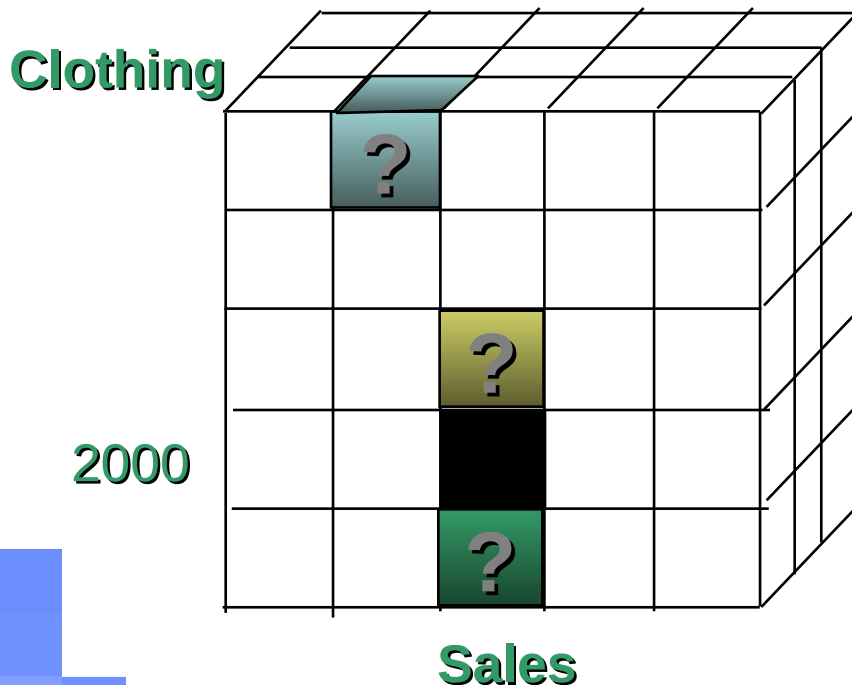


Naming Cells With Relative References...

(Products.Clothing, Measures.Sales, Time.[2000].PrevMember)

(Products.Clothing, Measures.Sales, Time.[2000].NextMember)

(Products.Clothing, Measures.Sales.PrevMember, Time.[2000].Lag(3))



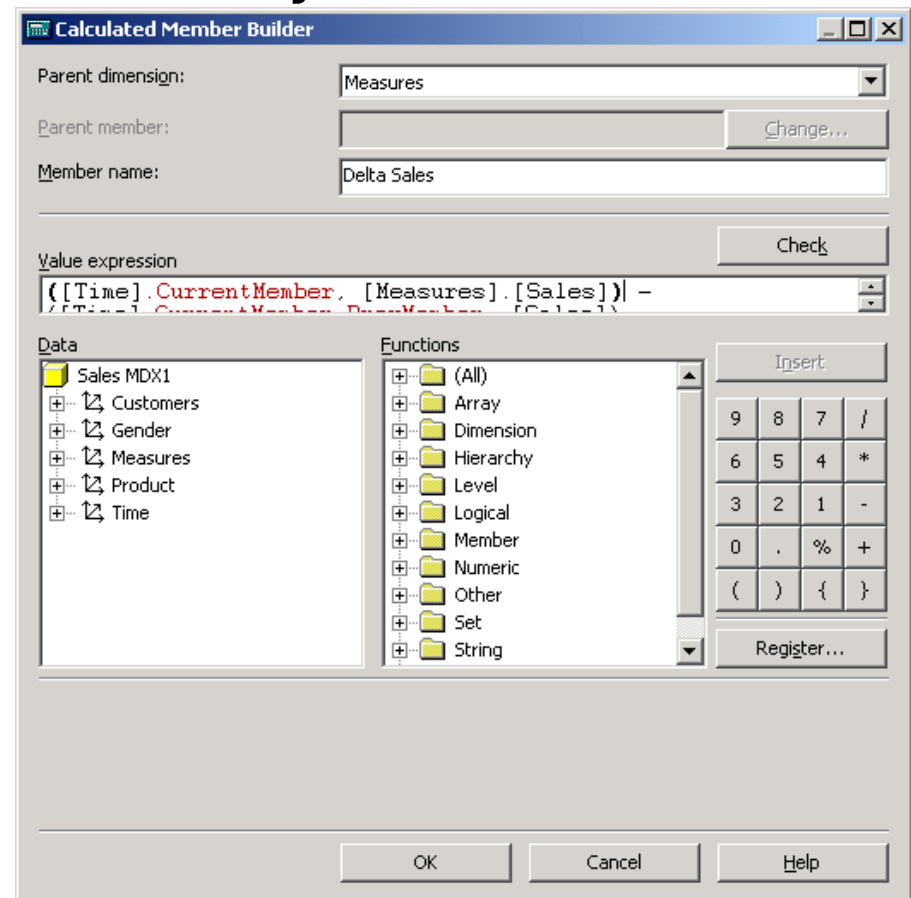
OR

(Products.Clothing,
Measures.Sales.PrevMember,
Time.[2000].Lead(-3))

Calculated Members

- Calculated members add significant power to Analysis cubes

- **Pre-define complex business logic**
- **Usually create calculated *measures***
- **Computed at run-time**



1. How Did Sales This Period Compare With Sales In The Previous Period?

Year	Qtr	Mon	Sales
2000			790
	Q1		120
		Jan	30
		Feb	40
		Mar	50
	Q2		200
		Apr	65
		May	45
		Jun	90
	Q3		185
		Jul	55
		Aug	60
		Sep	70
	Q4		285
		Oct	80
		Nov	100
		Dec	105

1. How Did Sales This Period Compare With Sales In The Previous Period?

Qtr	Mon	Sales	Delta
Q2		200	80
	Apr	65	15
	May	45	-20
	Jun	90	45

(Time.CurrentMember, Measures.CurrentMember) =

1. How Did Sales This Period Compare With Sales In The Previous Period?

Qtr	Mon	Sales	Delta
Q2		200	80
	Apr	65	15
	May	45	-20
	Jun	90	45

$(\text{Time.CurrentMember}, \text{Measures.CurrentMember}) =$

$(\text{Time.CurrentMember}, \text{Measures.Sales}) -$

1. How Did Sales This Period Compare With Sales In The Previous Period?

Qtr	Mon	Sales	Delta
Q2		200	80
	Apr	65	15
	May	45	-20
	Jun	90	45

(Time.CurrentMember, Measures.CurrentMember) =

(Time.CurrentMember, Measures.Sales)

-

(Time.CurrentMember.PrevMember, Measures.Sales)

Results For Question 1:

Year	Qtr	Mon	Sales	Delta
2000			790	
	Q1		120	
		Jan	30	
		Feb	40	10
		Mar	50	10
	Q2		200	80
		Apr	65	15
		May	45	-20
		Jun	90	45
	Q3		185	-15
		Jul	55	-35
		Aug	60	15
		Sep	70	10
	Q4		285	100
		Oct	80	10
		Nov	100	20
		Dec	105	5

2. How Did Sales In The Current Period Compare With Sales In The Same Period Last Year?

Year	Qtr	Mon	Sales
2000	Q1		790
			120
		Jan	30
		Feb	40
		Mar	50
	Q4		285
		Oct	80
		Nov	100
		Dec	105
2001	Q1		850
			170
		Jan	50
		Feb	55
		Mar	65
	Q4		275
		Oct	90
		Nov	100
		Dec	85

2. How Did Sales In The Current Period Compare With Sales In The Same Period Last Year?

Year	Qtr	Mon	Sales	Annual Delta
2000			790	
	Q1		120	
		Jan	30	
		Feb	40	
		Mar	50	
2001			850	
	Q1		170	
		Jan	50	20
		Feb	55	
		Mar	65	

(Time.CurrentMember, Measures.Sales)-

(Time.CurrentMember.Lag(12), Measures.Sales)

2. How Did Sales In The Current Period Compare With Sales In The Same Period Last Year?

Year	Qtr	Mon	Sales	Annual Delta
2000			790	
	Q1		120	
		Jan	30	
		Feb	40	
		Mar	50	
2001			850	
	Q1		170	
		Jan	50	20
		Feb	55	
		Mar	65	

(Time.CurrentMember, Measures.Sales)-

(Time.CurrentMember, Lag(12), Measures.Sales)

(ParallelPeriod(Year,1,Time.CurrentMember),
Measures.Sales)

Results For Question 2:

Year	Qtr	Mon	Sales	Ann. Delta
2000			790	
	Q1		120	
		Jan	30	
		Feb	40	
		Mar	50	
	Q4		285	
		Oct	80	
		Nov	100	
		Dec	105	
2001			850	60
	Q1		170	50
		Jan	50	20
		Feb	55	15
		Mar	65	15
	Q4		275	-10
		Oct	90	10
		Nov	100	0
		Dec	85	-20

3. What Have My Sales Been Since The Beginning Of The Year?

Year	Qtr	Mon	Sales
2000			790
	Q1		120
		Jan	30
		Feb	40
		Mar	50
	Q2		200
		Apr	65
		May	45
		Jun	90
	Q3		185
		Jul	55
		Aug	60
		Sep	70
	Q4		285
		Oct	80
		Nov	100
		Dec	105

3. What Have Sales Been Since The Beginning Of The Year?

Year	Qtr	Mon	Sales	YTD
2000			790	
	Q1		120	120
		Jan	30	30
		Feb	40	70
		Mar	50	120

Time.Jan, Measures.Sales +

Time.Feb, Measures.Sales +

Time.Mar, Measures.Sales ...

3. What Have Sales Been Since The Beginning Of The Year?

Year	Qtr	Mon	Sales	YTD
2000			790	
	Q1		120	120
		Jan	30	30
		Feb	40	70
		Mar	50	120

Time.Jan, Measures.Sales +

Time.Feb, Measures.Sales +

Time.Mar, Measures.Sales =

NO!

Sum(YTD(Time.CurrentMember), Sales)

Results For Question 3:

Year	Qtr	Mon	Sales	YTD
2000			790	
	Q1		120	120
		Jan	30	30
		Feb	40	70
		Mar	50	120
	Q2		200	320
		Apr	65	185
		May	45	230
		Jun	90	320
	Q3		185	505
		Jul	55	375
		Aug	60	435
		Sep	70	505
	Q4		285	790
		Oct	80	585
		Nov	100	685
		Dec	105	790

Questions